

Meter To MEASURE Short DISTANCES

ARNAV BANSAL

This Arduino-based distance meter can be used for measuring the distance between the sensor and object in the range of 2cm

cost ultrasonic sensor that consists of transmitter and receiver units.

When the controller gives a high signal to trig pin of the sensor, the

sensor emits eight-cycle sonic bursts at 40kHz. After sending the sonic bursts, the sensor sets the echo to high till the sonic bursts return to the sensor. The length travelled by this pulse shows how far the object is.

Ultrasonic waves follow the properties of generic sound waves. Due to this, there might be a change (microscopic) in speed of sound waves in either direction with change in temperature. This change generates errors in calculation and, hence, different values are obtained for the same object kept at the same place when the temperature changes. To overcome this, a special algorithm is used, which gives the exact distance. This algorithm is based on probability and statistics.

Software

The software used to program Arduino Uno is Arduino IDE. It can be downloaded for free from www.arduino.cc/en/Main/Software.

The open source Arduino software (IDE) makes it easy to write codes and upload to the board.

The library used in this project is Ultrasonic Library, which can be downloaded from <https://github.com/arnavbansal95/Ultrasonic-Arduino-HC-SR04/archive/master.zip>

To install this library, extract the downloaded folder, copy

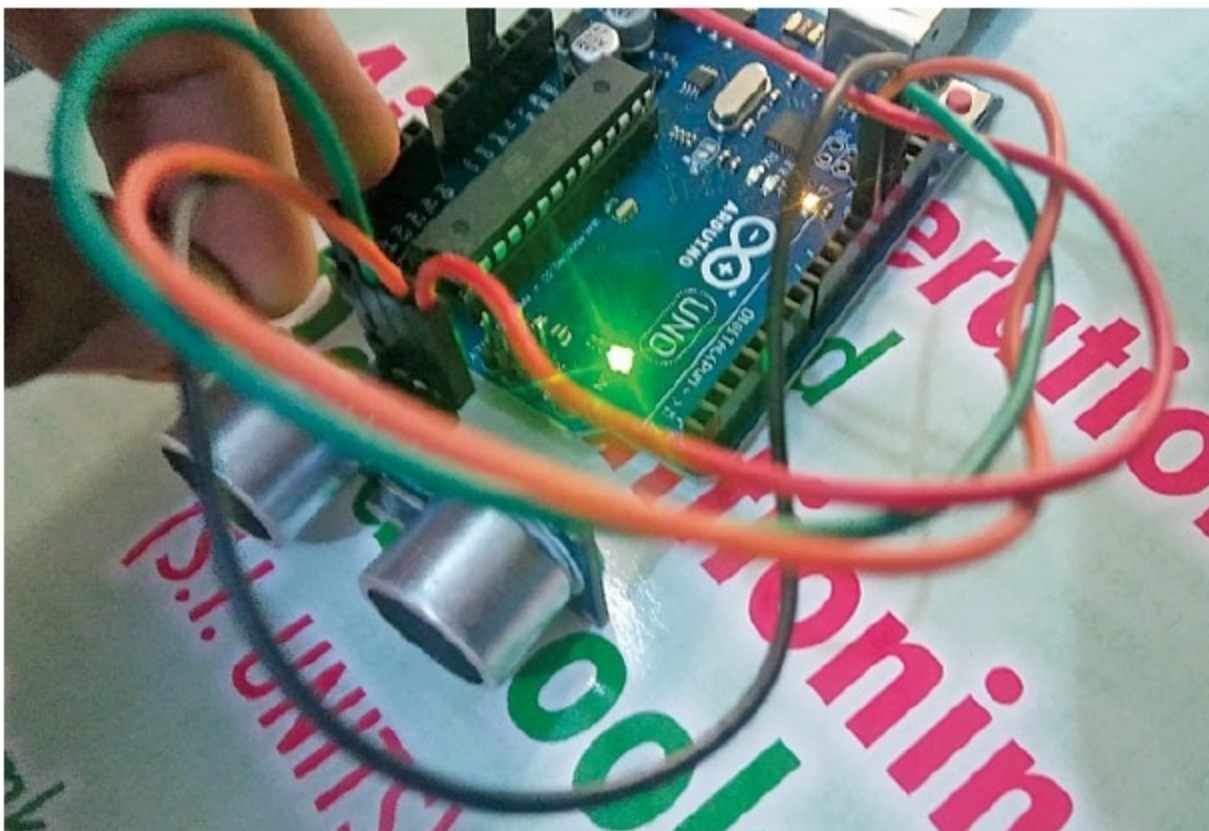


Fig. 1: Author's prototype

to 400cm using an ultrasonic sensor. The author's prototype and ultrasonic module are shown in Figs 1 and 2, respectively.

Circuit and working

The circuit diagram of the distance meter is shown in Fig. 3. It is built around Arduino Uno board (board1), HC-SR04 ultrasonic sensor and a few other components. HC-SR04 is a low-



Fig. 2: Ultrasonic sensor module

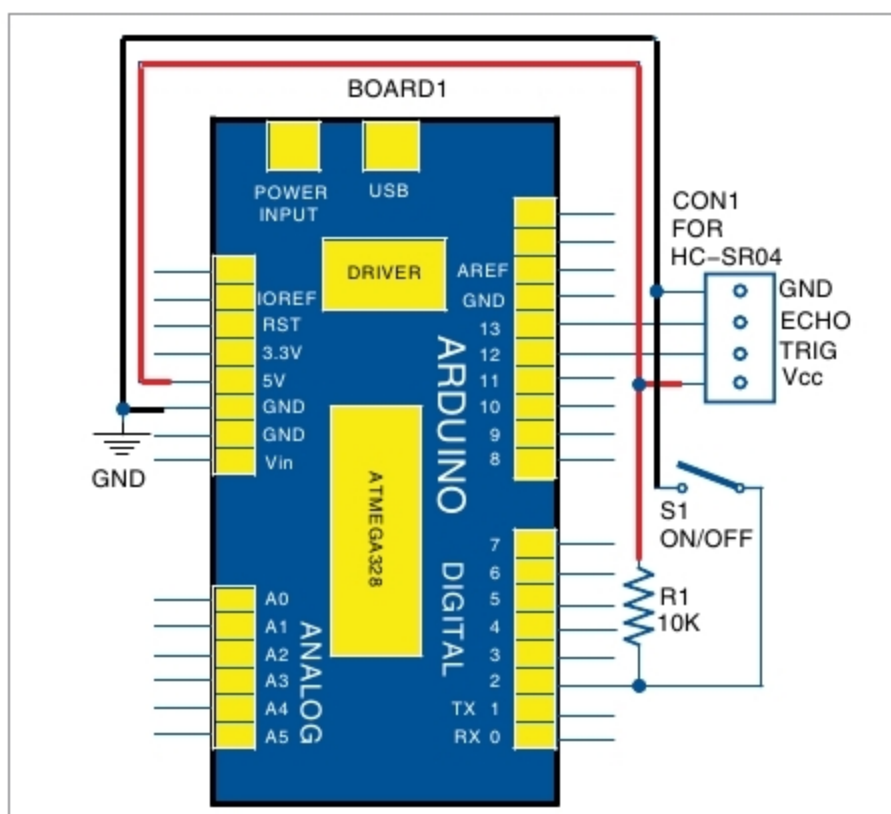


Fig. 3: Circuit diagram of Arduino-based distance meter